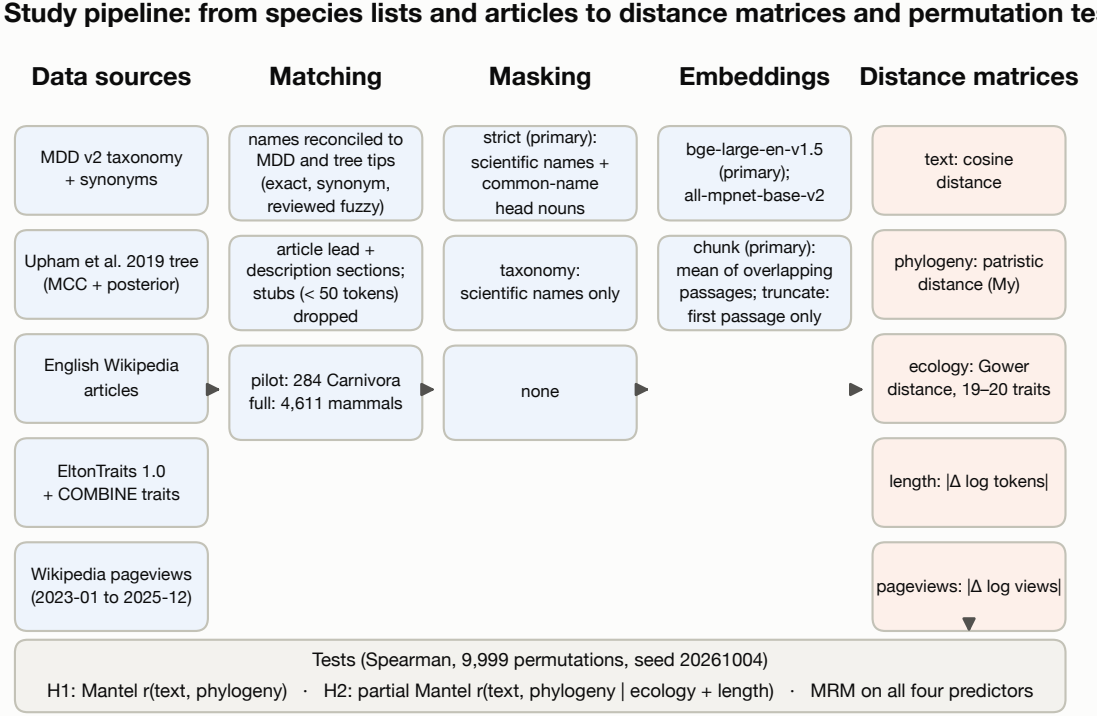
Study pipeline: from species lists and articles to distance matrices and permutation tests
Data sources
Matching
Masking
Embeddings
Distance matrices
MDD v2 taxonomy + synonyms
Upham et al. 2019 tree (MCC + posterior)
English Wikipedia articles
EltonTraits 1.0 + COMBINE traits
Wikipedia pageviews (2023-01 to 2025-12)
names reconciled to MDD and tree tips (exact, synonym, reviewed fuzzy)
article lead + description sections; stubs (< 50 tokens) dropped
pilot: 284 Carnivora full: 4,611 mammals
strict (primary): scientific names + common-name head nouns
taxonomy: scientific names only
none
bge-large-en-v1.5 (primary); all-mpnet-base-v2
chunk (primary): mean of overlapping passages; truncate: first passage only
text: cosine distance
phylogeny: patristic distance (My)
ecology: Gower distance, 19–20 traits
length: |Δ log tokens|
pageviews: |Δ log views|
Tests (Spearman, 9,999 permutations, seed 20261004)
H1: Mantel r(text, phylogeny) · H2: partial Mantel r(text, phylogeny | ecology + length) · MRM on all four predictors